

CSE 311:

Data Communication

Instructor:

Dr. Md. Monirul Islam

Analysis of Signals and Systems

All Pass Vs. Distortionless System

Distortionless system:

$$|H(f)| = k \quad \text{and} \quad \theta_h(f) = -2\pi f t_d$$

- Linear phase $\theta_h(f) = -2\pi f t_d$ guarantees a constant time delay t_d for every sinusoid
- Synchronization is guaranteed

All Pass Vs. Distortionless System

How to Check Distortionless transmission?

Find the slope (derivative) of $\theta_h(f)$

$$t_d(f) = -\frac{1}{2\pi} \cdot \frac{d\theta_h(f)}{df}$$

- If the slope of $\theta_h(f)$ is constant, it guarantees a constant time delay t_d for every sinusoid
- If the slope is a function of f , different frequency components will have different time delay

Nature of Distortion in Audio Signals

Human ear is

- **sensitive** to amplitude distortion which results if $|H(f)| \neq k$
- **insensitive** to phase distortion which results if the slope of $\theta_h(f)$ is NOT constant

Nature of Distortion in Audio Signals

- Duration of each spoken syllable : 0.01 to 0.1 second
- Variation of the slope of $\theta_h(f)$ is ONLY fraction of a millisecond, meaning a nearly constant slope, e.g., time delay

Nature of Distortion in Video Signals

Human eye is

- insensitive to amplitude distortion
- sensitive to phase distortion

Nature of Distortion in Video Signals

Insensitive amplitude distortion

- amplitude distortion changes only half-tone values of picture

Sensitive phase distortion

- Nonlinear phase distortion means different time delay for different picture (signal) element
- This results in smeared picture

Nature of Distortion in Communication Systems

Sensitive phase distortion

- Nonlinear phase distortion causes **pulse dispersion** (spreading out)
- **Pulses are interfered** with near by pulses
- This results in misinterpretation in the receiver
 - Binary **1** may read as **0** and vice versa

Ideal Vs. Practical Filters

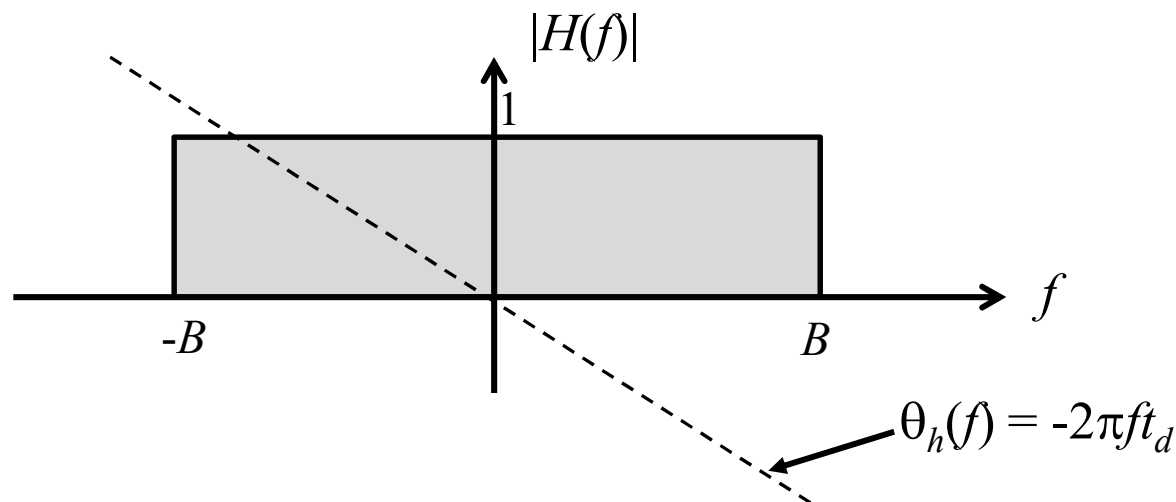
Ideal filters

- allow distortionless transmission for certain band of frequencies
- suppress all other frequencies

Ideal Vs. Practical Filters

This Ideal Lowpass filter

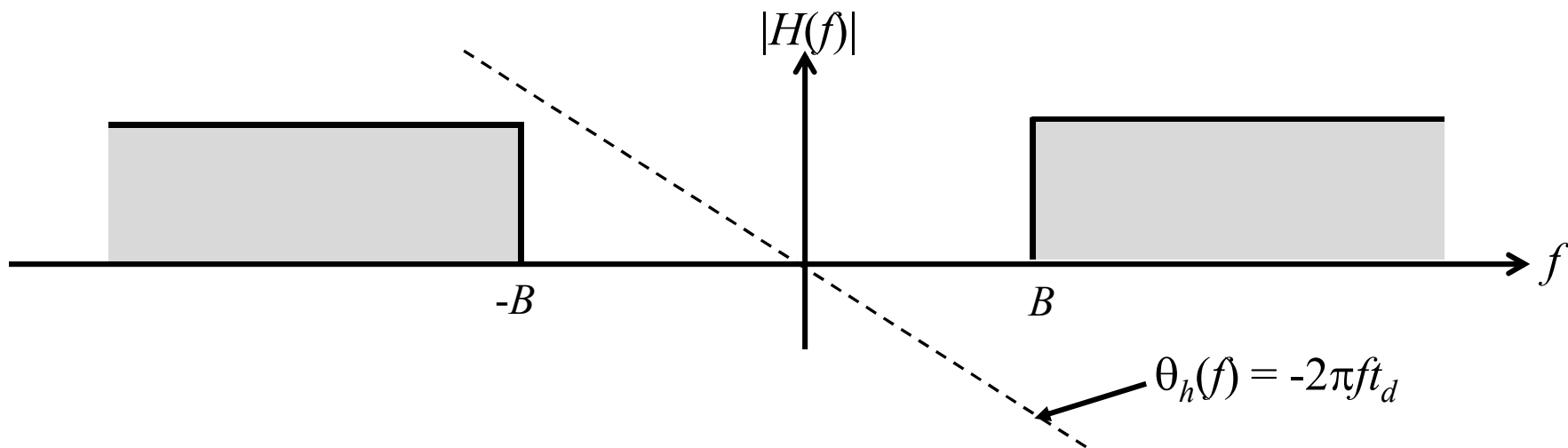
- allows transmission for frequencies $f \leq |B|$
- suppresses all other frequencies $f \geq |B|$



Ideal Vs. Practical Filters

This Ideal Highpass filter

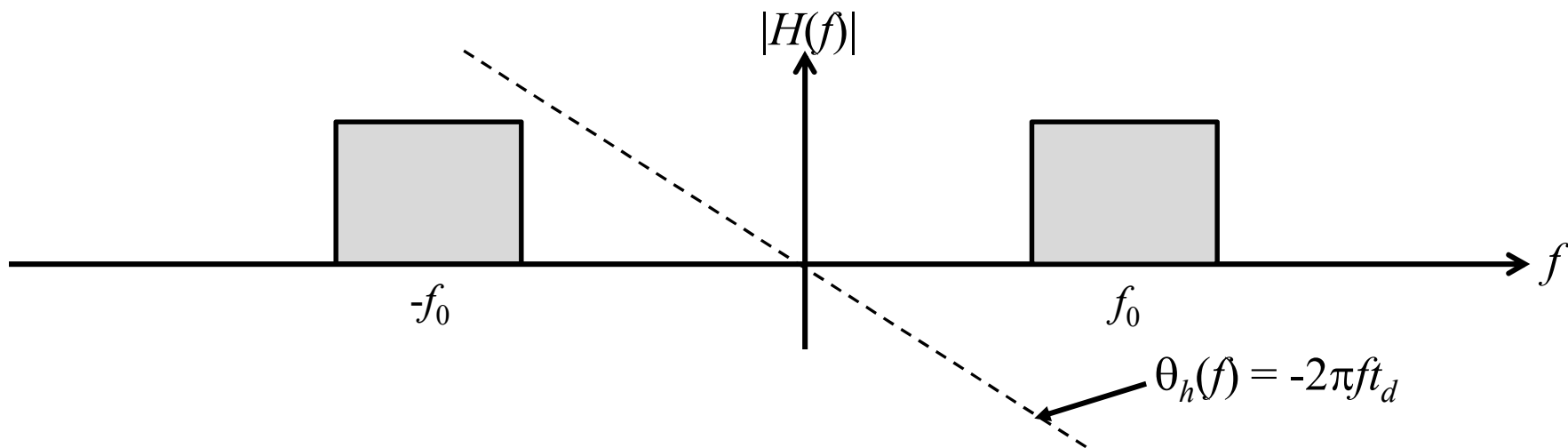
- allows transmission for frequencies $f \geq |B|$
- suppresses all other frequencies $f \leq |B|$



Ideal Vs. Practical Filters

This Ideal Bandpass filter

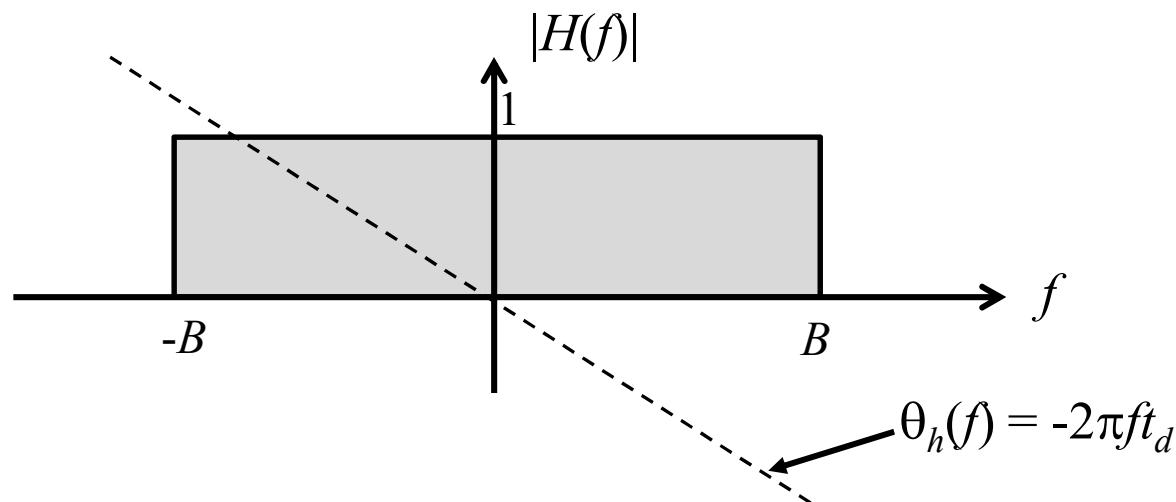
- allows transmission for the band centered at $\pm f_0$
- suppresses all other frequencies



Ideal Vs. Practical Filters

This Ideal Lowpass filter

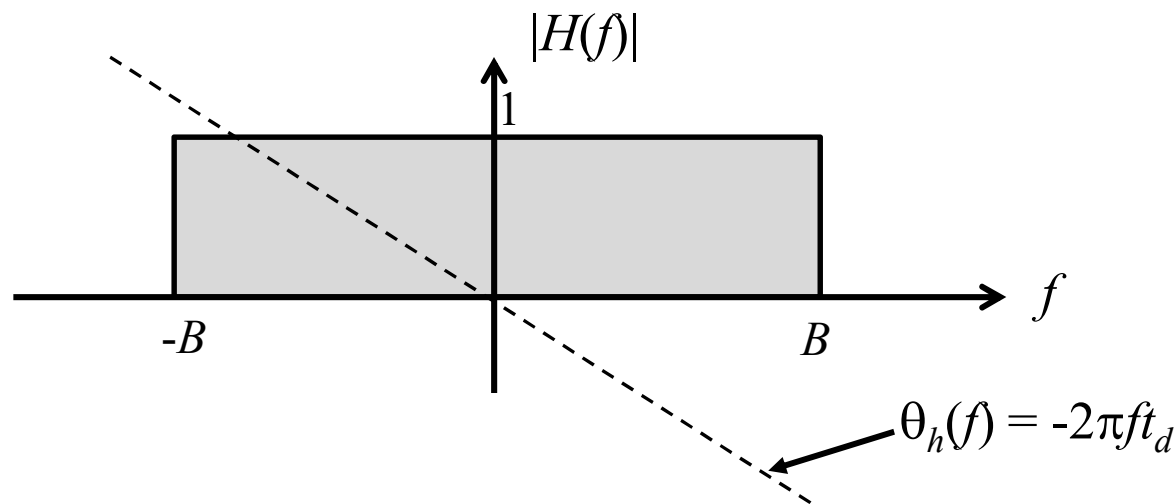
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Ideal Vs. Practical Filters

This Ideal Lowpass filter

- allows transmission for frequencies $f \leq |B|$
- suppresses all other frequencies $f \geq |B|$
- Slope of $\theta_h(f)$ is constant $-t_d$, means time delay is t_d

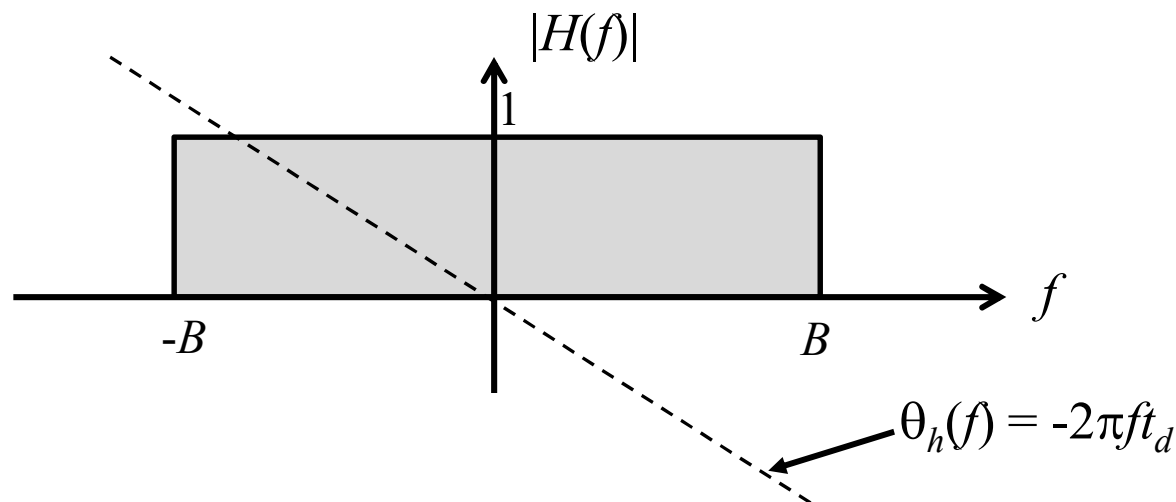


Ideal Vs. Practical Filters

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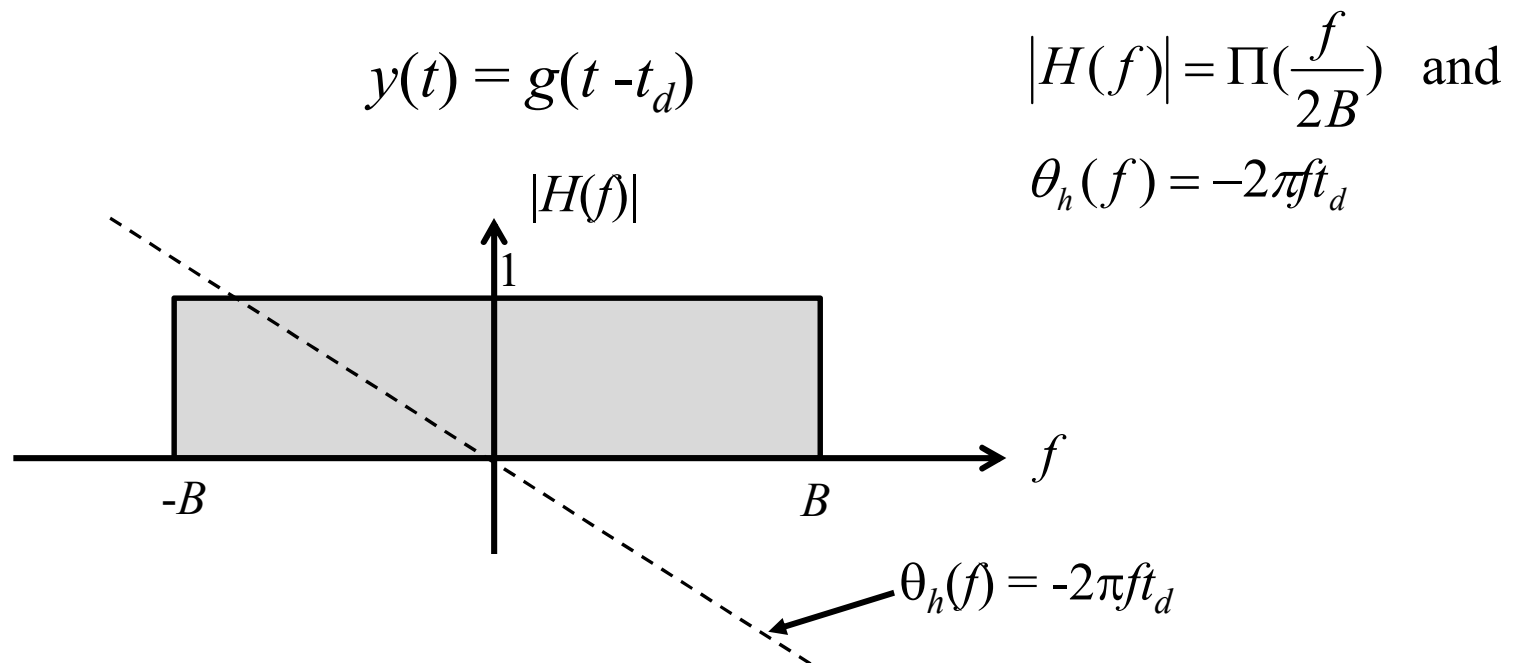
$$y(t) = g(t - t_d)$$



Ideal Vs. Practical Filters

This Ideal Lowpass filter

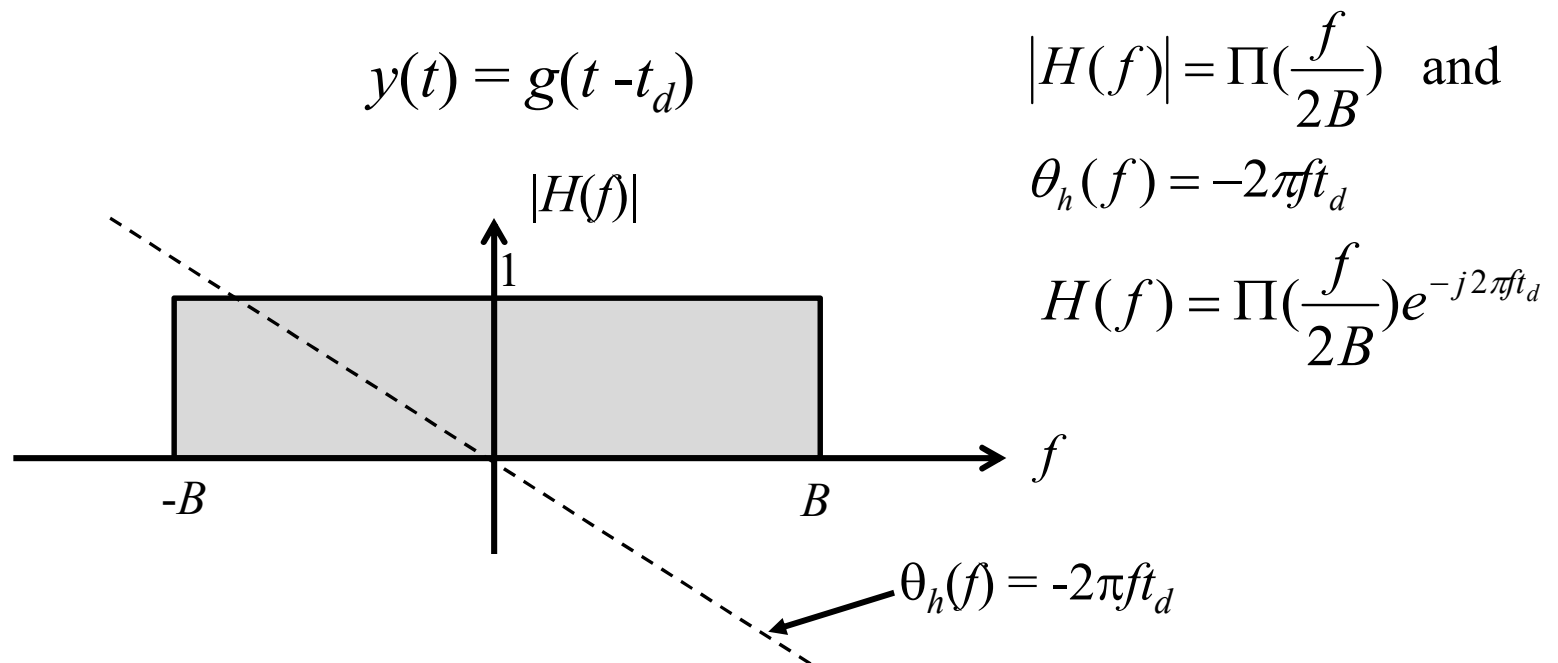
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Ideal Vs. Practical Filters

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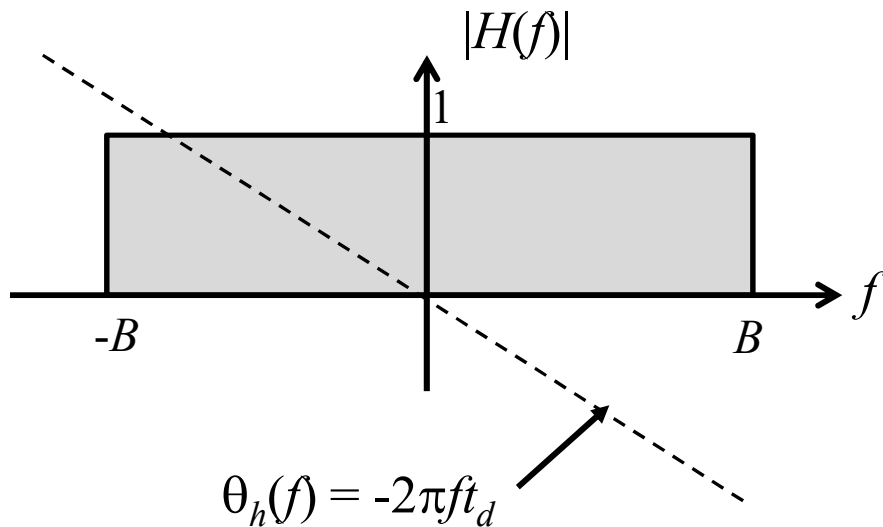


Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter

Frequency Transform of
Impulse response

$$H(f) = \Pi\left(\frac{f}{2B}\right)e^{-j2\pi ft_d}$$



Ideal Vs. Practical Filters

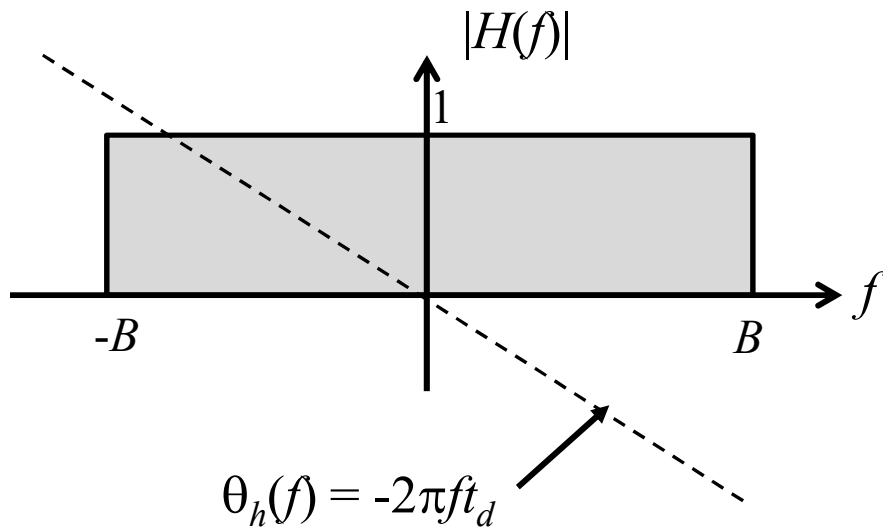
Impulse response of Ideal Lowpass filter

Frequency Transform of
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$$H(f) = \Pi\left(\frac{f}{2B}\right)e^{-j2\pi ft_d}$$

Impulse response

$$\begin{aligned} h(t) &= \mathfrak{F}^{-1}[H(f)] = \mathfrak{F}^{-1}\left[\Pi\left(\frac{f}{2B}\right)e^{-j2\pi ft_d}\right] \\ &= 2B\text{sinc}(2\pi B(t - t_d)) \end{aligned}$$

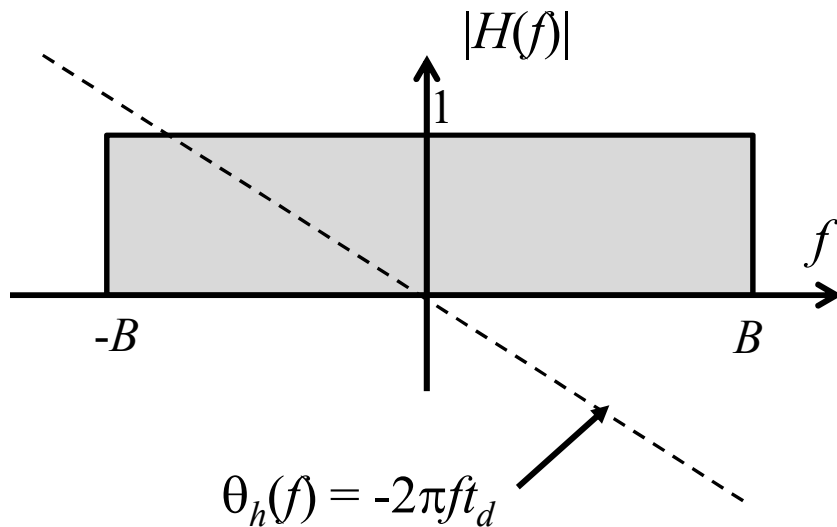


Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter

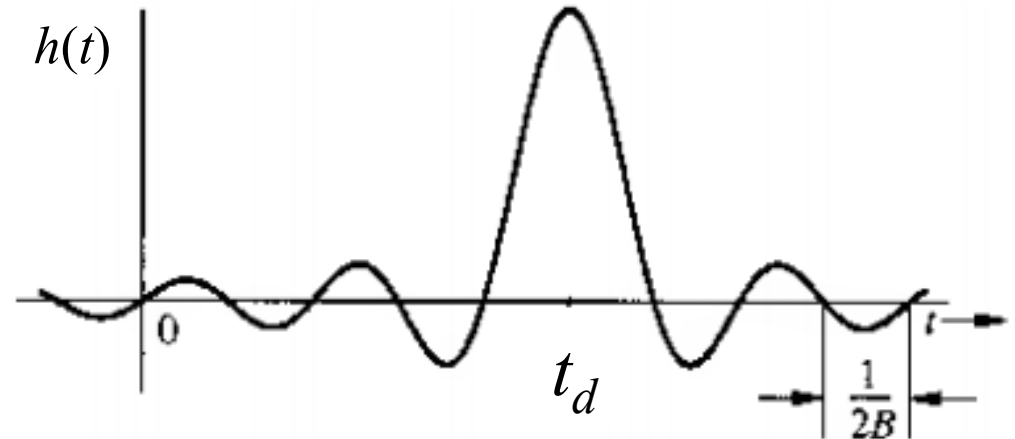
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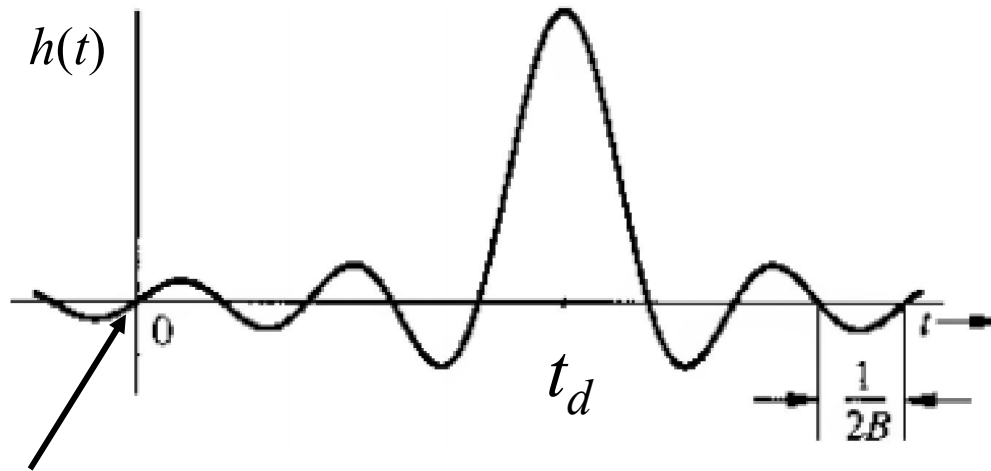
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Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter

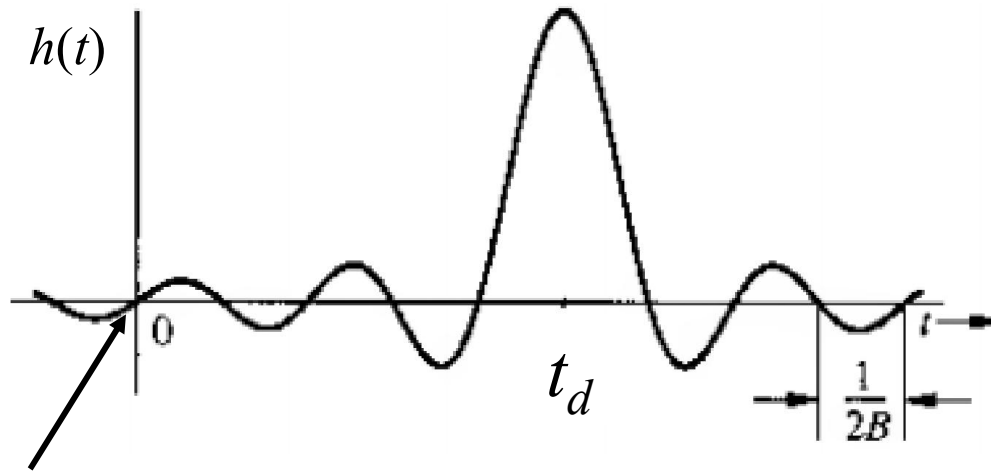


Impulse response
 $h(t) = 2B \text{sinc}(2\pi B(t - t_d))$

Response begins
before $t = 0$

Ideal Vs. Practical Filters

Impulse response of Ideal Lowpass filter



Impulse response

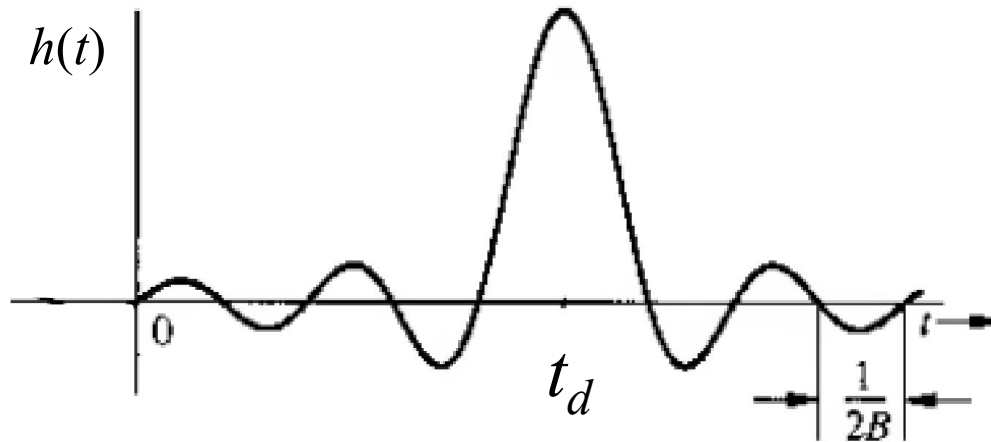
$$h(t) = 2B \text{sinc}(2\pi B(t - t_d))$$

Response begins
before $t = 0$

Noncausal and
unrealizable filter

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



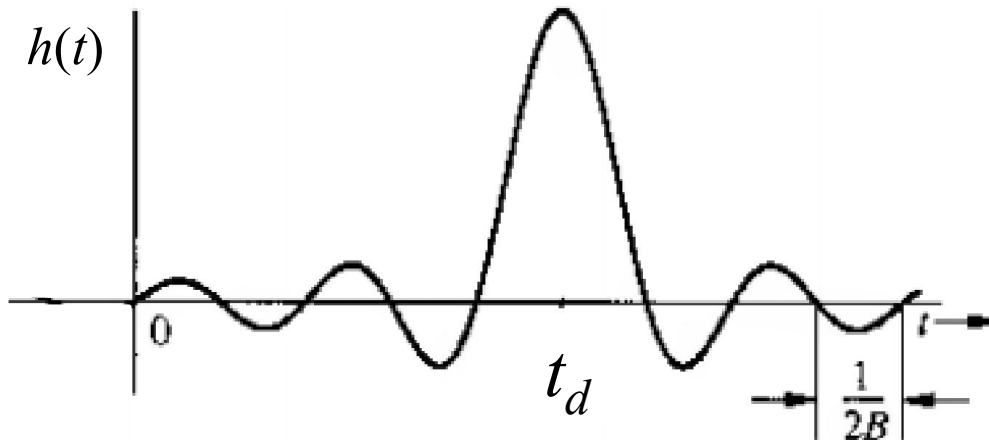
Condition in time domain

$$h(t) = 0 \text{ for all } t < 0$$

Response begins
on or after $t = 0$

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



Response begins
on or after $t = 0$

Condition in time domain

$$h(t) = 0 \text{ for all } t < 0$$

Condition in frequency
domain

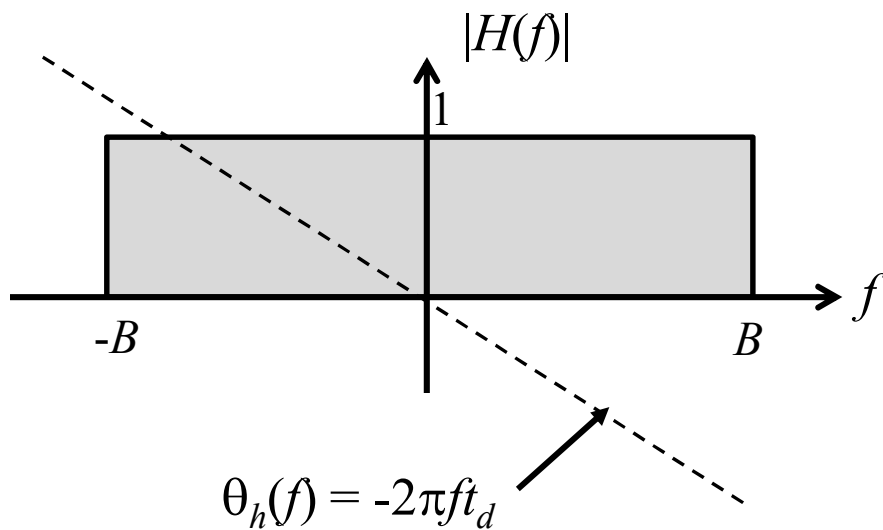
$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty$$

$$\text{where } \alpha(f) = \ln|H(f)|$$

Paley-Wiener Criterion

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



Condition in frequency domain

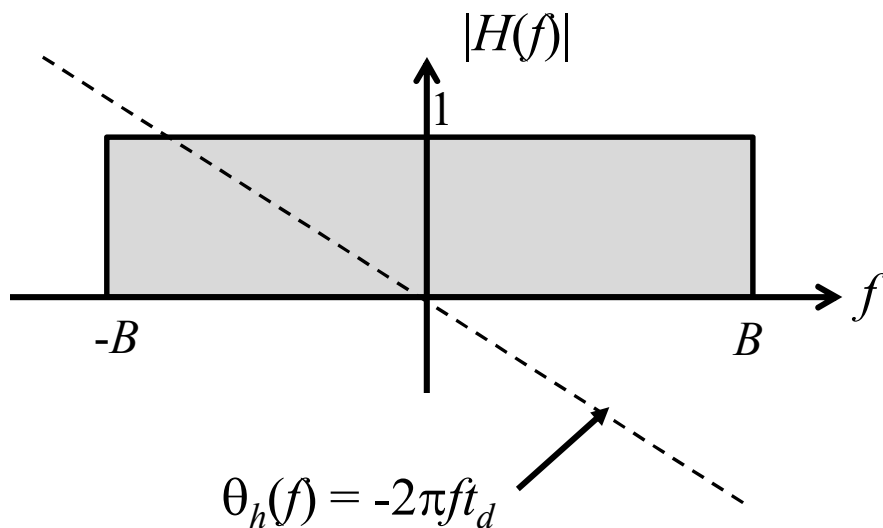
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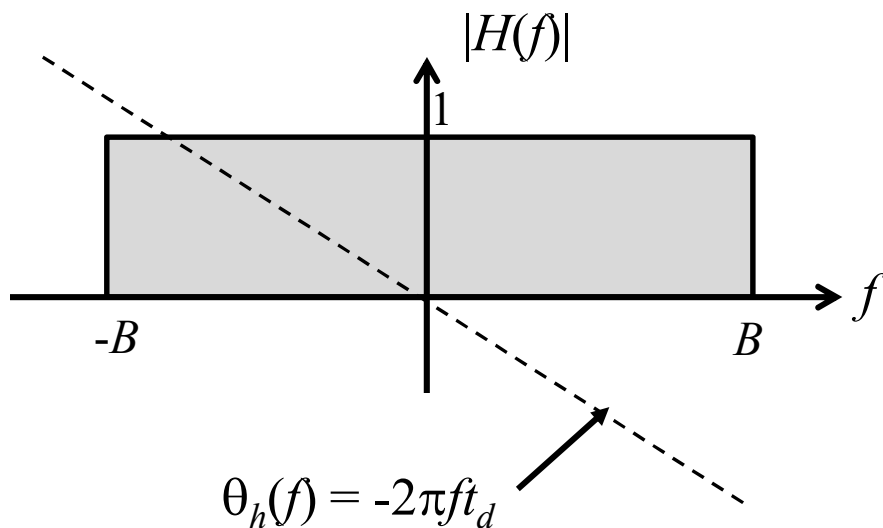
where $\alpha(f) = \ln |H(f)|$

Paley-Wiener Criterion

- If for **certain band** of frequencies $|H(f)| = 0$ means $|\alpha(f)| = \ln |H(f)| = \infty$, which does **NOT** satisfy P-W criterion

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



Condition in frequency domain

$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty$$

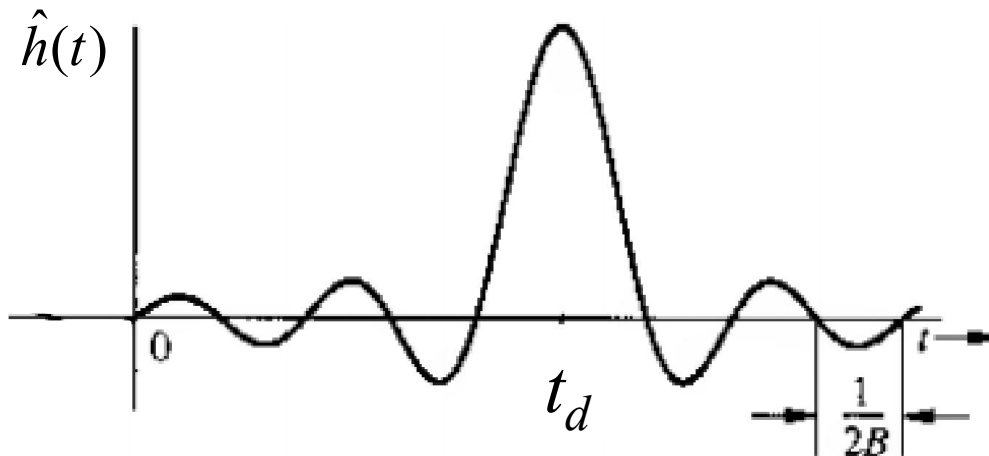
where $\alpha(f) = \ln|H(f)|$

Paley-Wiener Criterion

- If for **discrete** of frequencies $|H(f)| = 0$, P-W criterion may be satisfied

Ideal Vs. Practical Filters

Causal and realizable Lowpass filter



$$\hat{h}(t) = h(t)u(t)$$

Condition in frequency domain

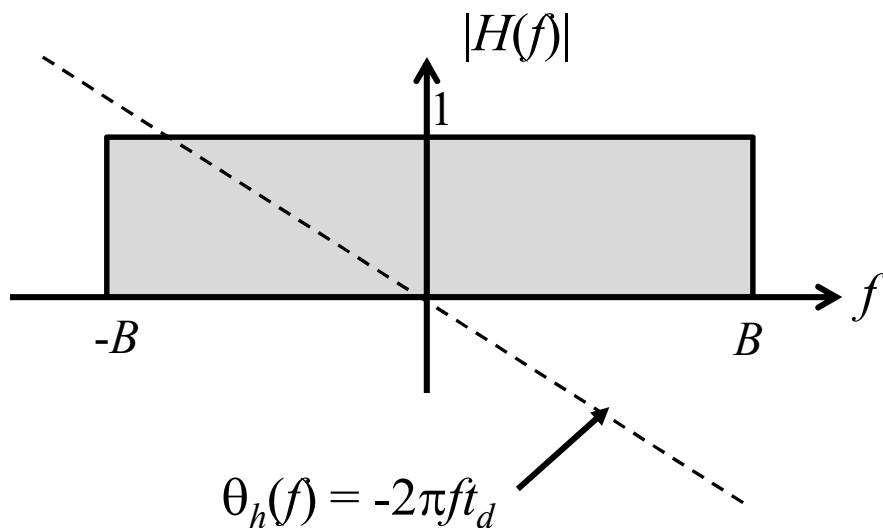
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Paley-Wiener Criterion

Ideal Vs. Practical Filters

Other Causal and realizable Lowpass filters



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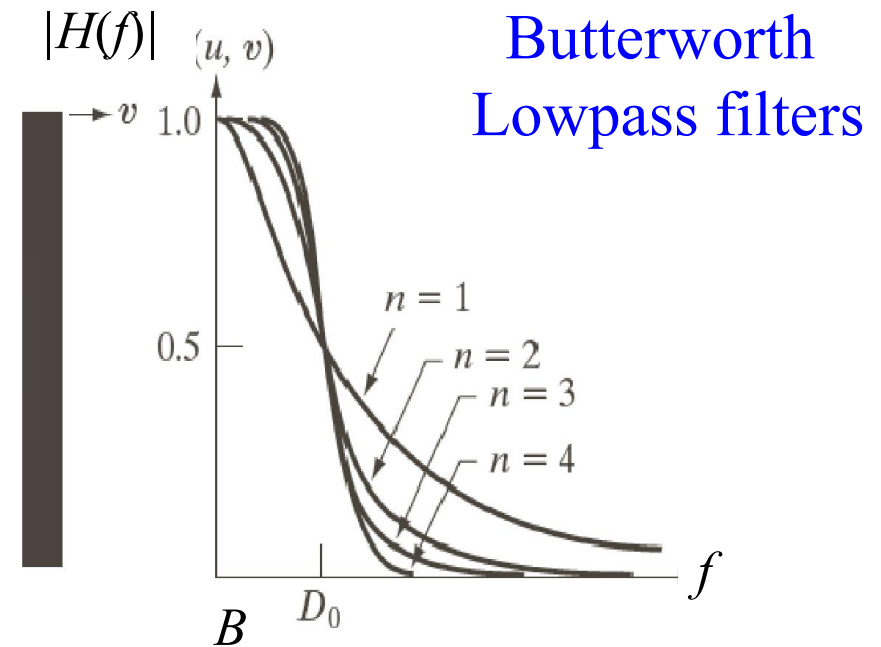
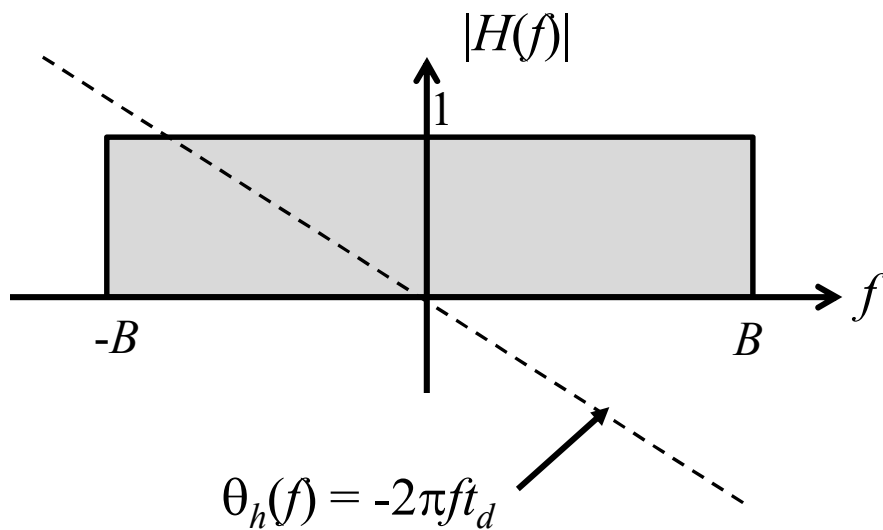
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Paley-Wiener Criterion

- Instead of sharp change at $\neq B$, change smoothly and approach asymptotically to zero

Ideal Vs. Practical Filters

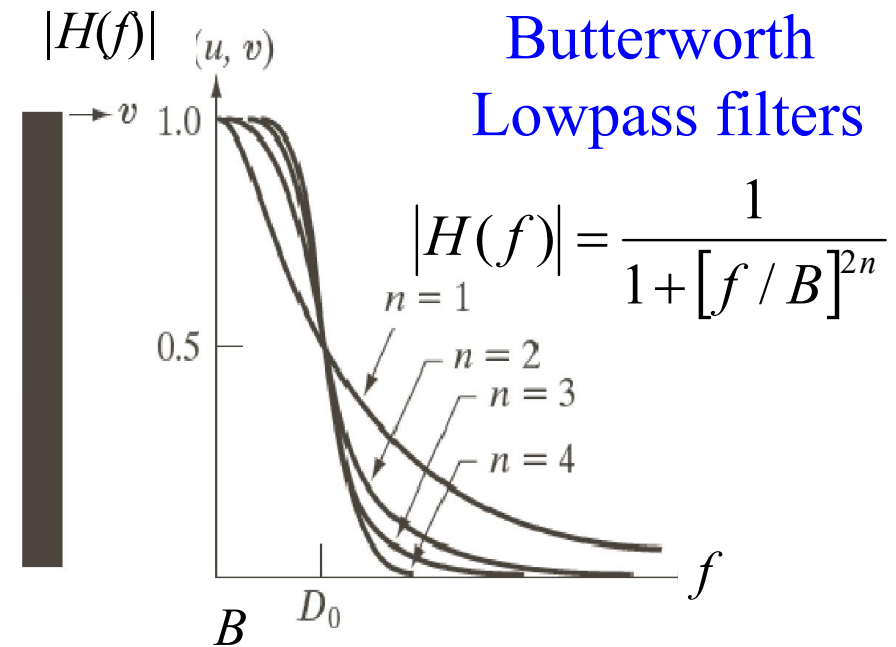
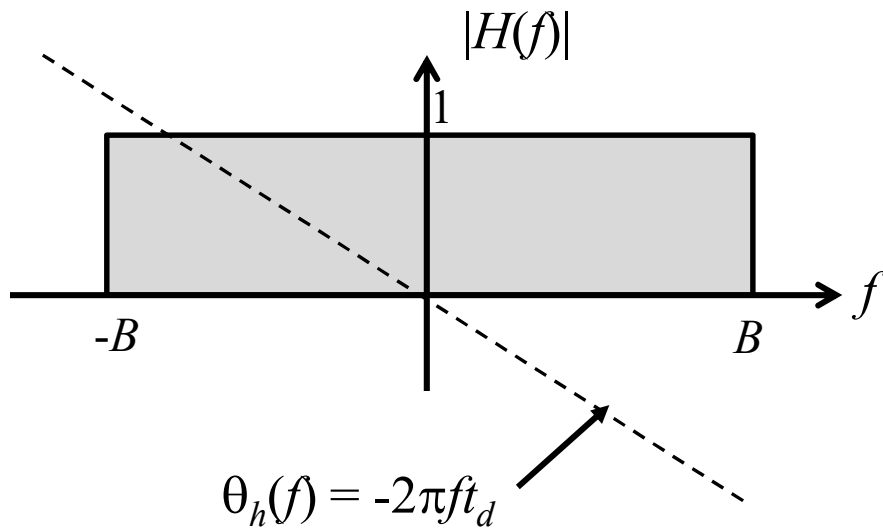
Other Causal and realizable Lowpass filters



- Instead of sharp change at $\pm B$, change smoothly and approach asymptotically to zero

Ideal Vs. Practical Filters

Other Causal and realizable Lowpass filters



- Instead of sharp change at $\pm B$, change smoothly and approach asymptotically to zero

Parseval's Theorem: Signal Energy in Frequency Domain

From Chapter 2,

Energy for real signal,

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt$$

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For complex valued signal,

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Parseval's Theorem: Signal Energy in Frequency Domain

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Energy for real signal,

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For complex valued signal,

$$\begin{aligned} E_g &= \int_{-\infty}^{\infty} |g(t)|^2 dt \\ &= \int_{-\infty}^{\infty} g(t) g(t)^* dt \end{aligned}$$

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$$\begin{aligned} g(t) &= \int_{-\infty}^{\infty} G(f)e^{j2\pi ft} df \\ g^*(t) &= \int_{-\infty}^{\infty} G(f)^* |e^{j2\pi ft}|^* df \\ &= \int_{-\infty}^{\infty} G(f)^* e^{-j2\pi ft} df \end{aligned}$$

Parseval's Theorem: Signal Energy in Frequency Domain

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} g(t)g(t)^* dt$$

$$= \int_{-\infty}^{\infty} g(t) \left[\int_{-\infty}^{\infty} G(f)^* e^{-j2\pi ft} df \right] dt$$

Parseval's Theorem: Signal Energy in Frequency Domain

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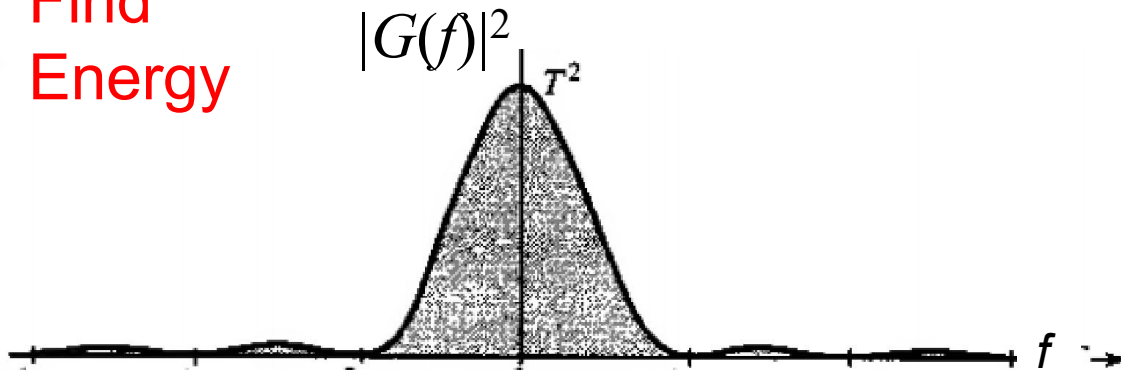
Parseval's Theorem: Signal Energy in Frequency Domain

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt = \int_{-\infty}^{\infty} |G(f)|^2 df$$

Application of Parseval's Theorem

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt = \int_{-\infty}^{\infty} |G(f)|^2 df$$

Find
Energy

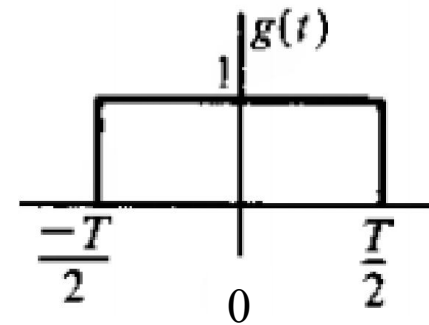
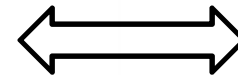
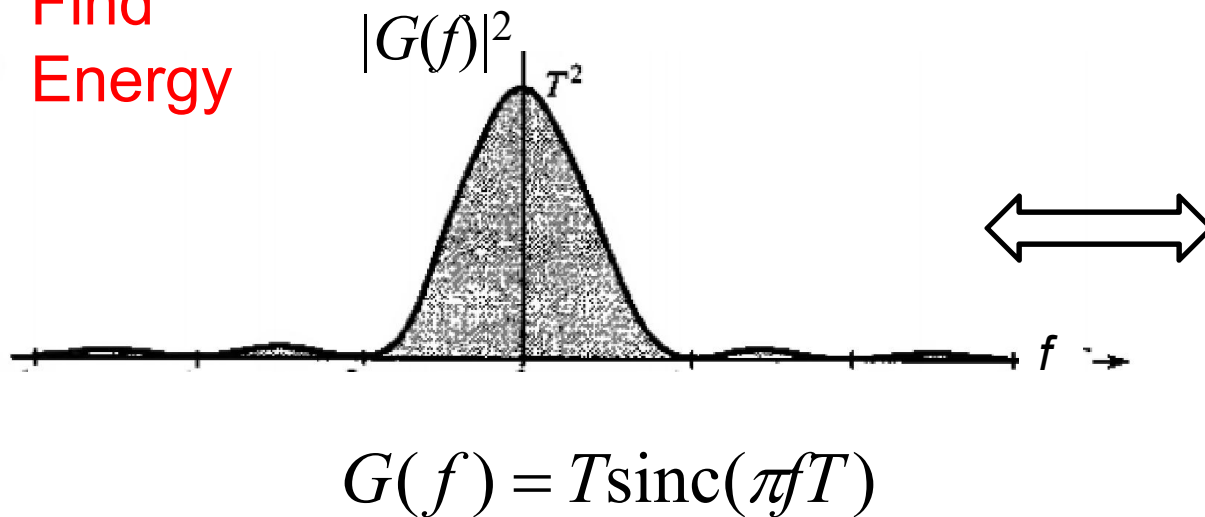


$$G(f) = T \text{sinc}(\pi f T)$$

Application of Parseval's Theorem

$$E_g = \int_{-\infty}^{\infty} |g(t)|^2 dt = \int_{-\infty}^{\infty} |G(f)|^2 df$$

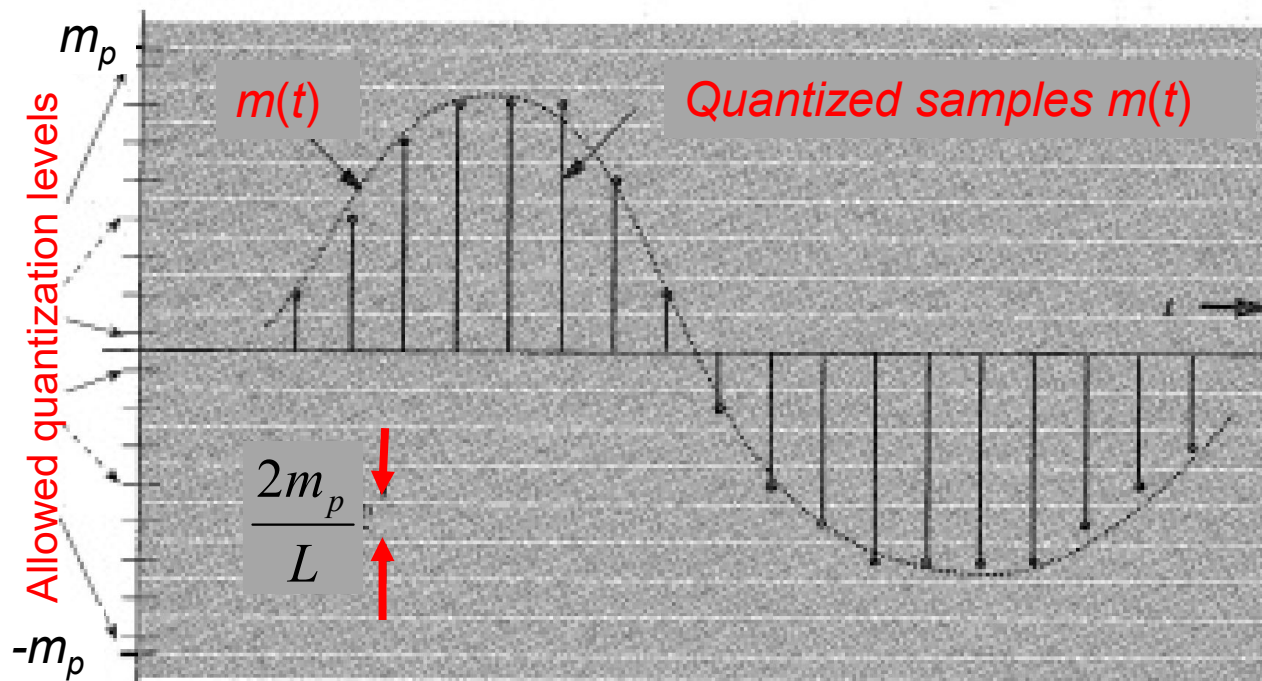
Find
Energy



Sampling and Analog-to-Digital Conversion

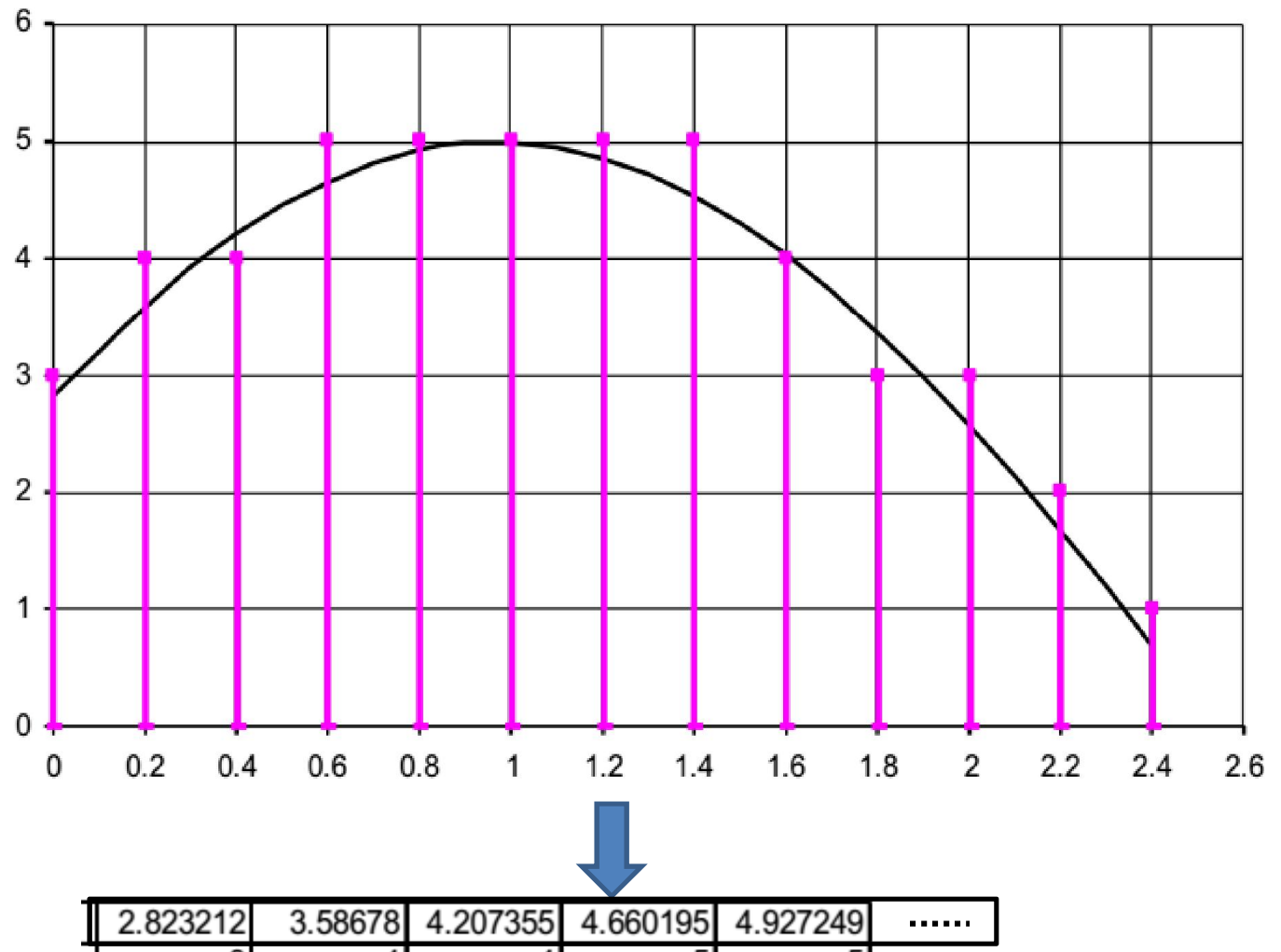
Analog to Digital Conversion of Message Signal

- 2 major steps
 - Sampling
 - Quantization



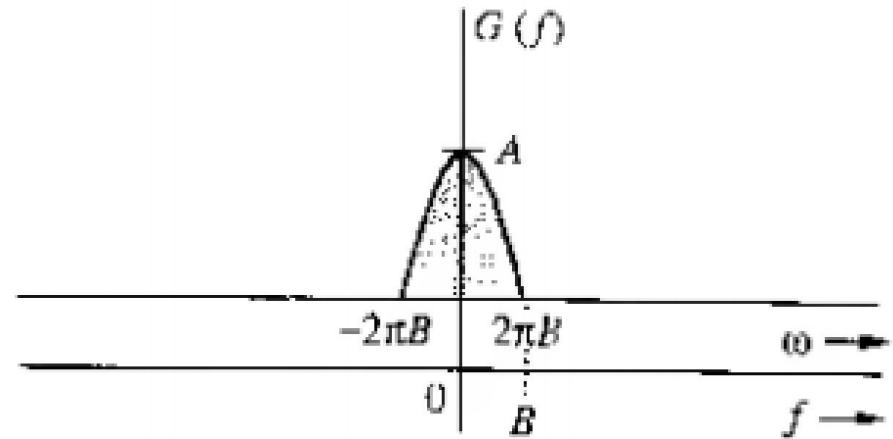
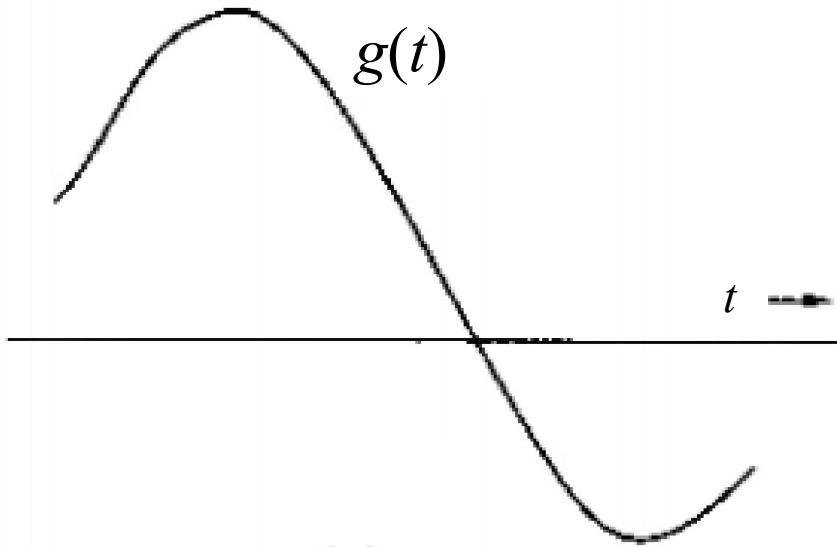
Analog to Digital Conversion of Message Signal

- Sampling



Sampling Theorem

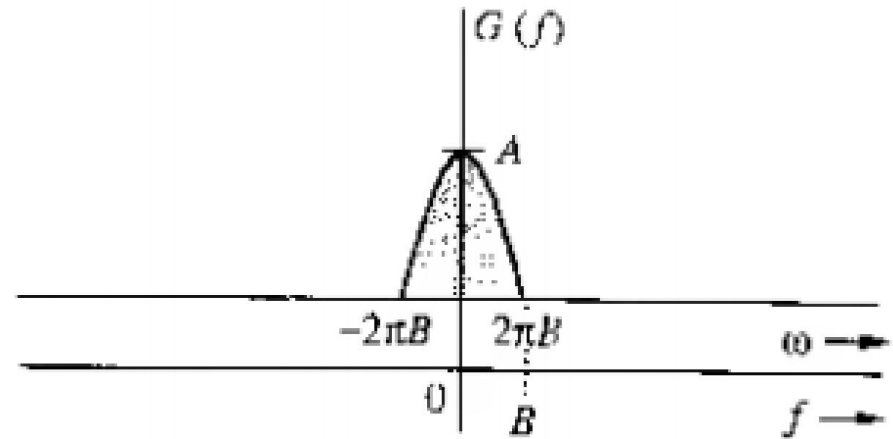
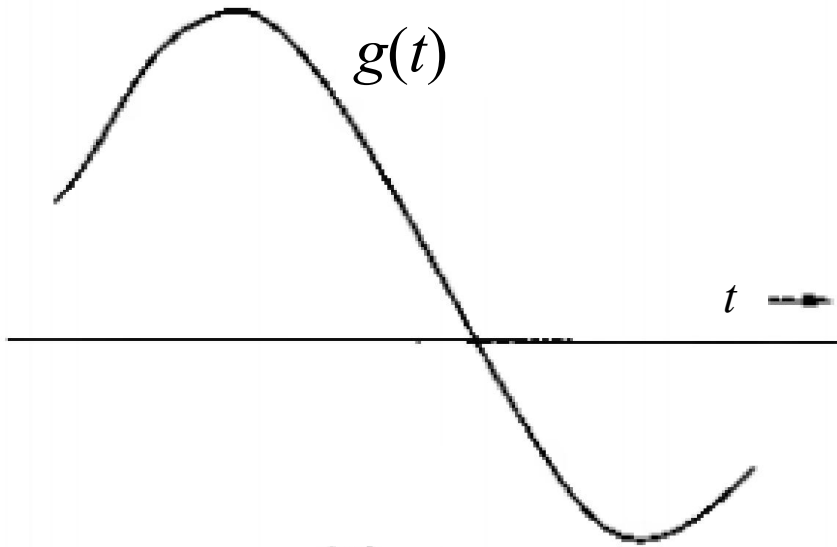
Let a message signal $g(t)$ band limited to B Hz



$$G(f) = 0 \text{ for all } |f| > B$$

Sampling Theorem

Let a message signal $g(t)$ band limited to B Hz

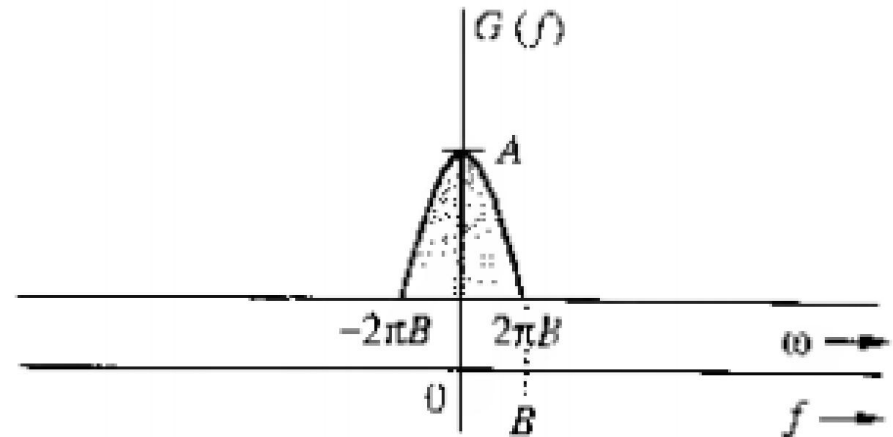
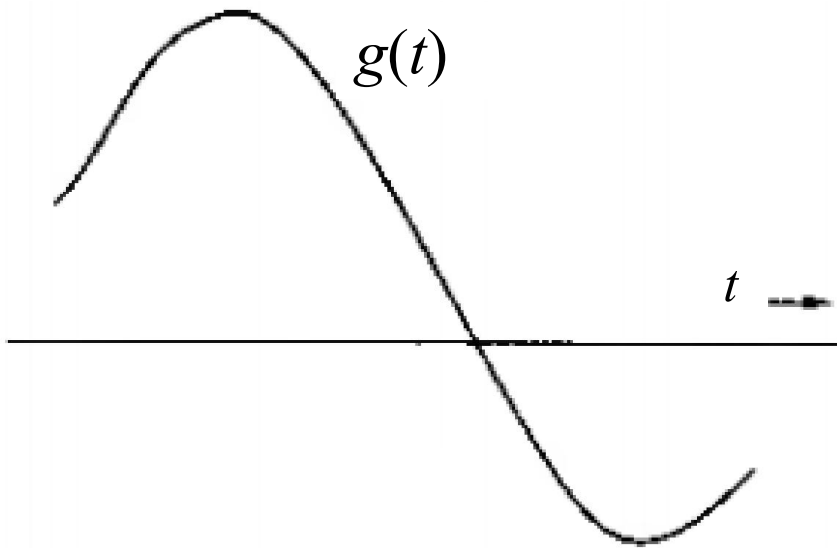


$$G(f) = 0 \text{ for all } |f| > B$$

The signal $g(t)$ can be reconstructed exactly from its discrete samples taken at a rate of f_s Hz, where $f_s \geq 2B$

Sampling Theorem

Let a message signal $g(t)$ band limited to B Hz

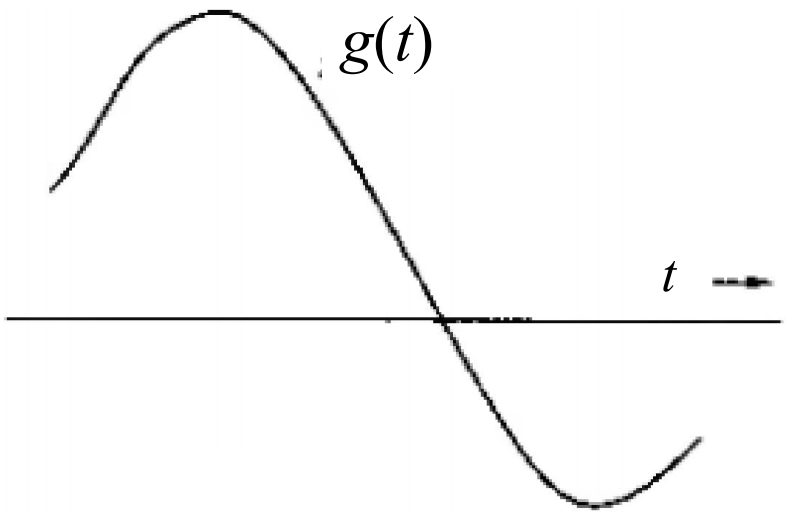


$$G(f) = 0 \text{ for all } |f| > B$$

For recovery, the minimum sampling $f_s = 2B$ Hz

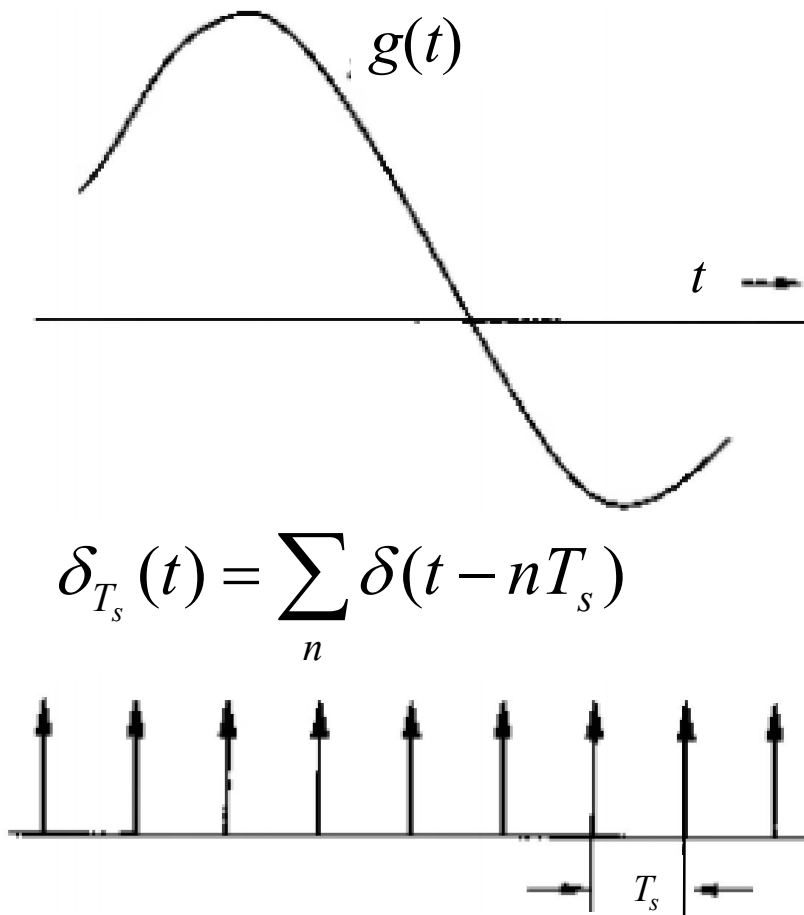
Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s



Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s



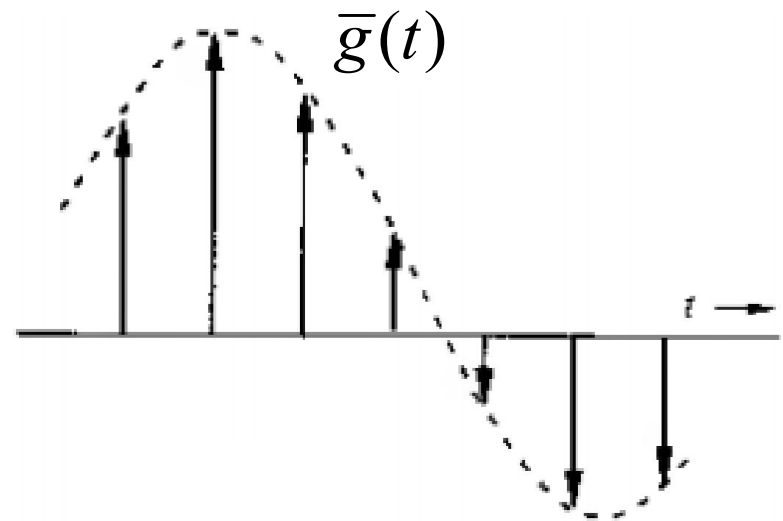
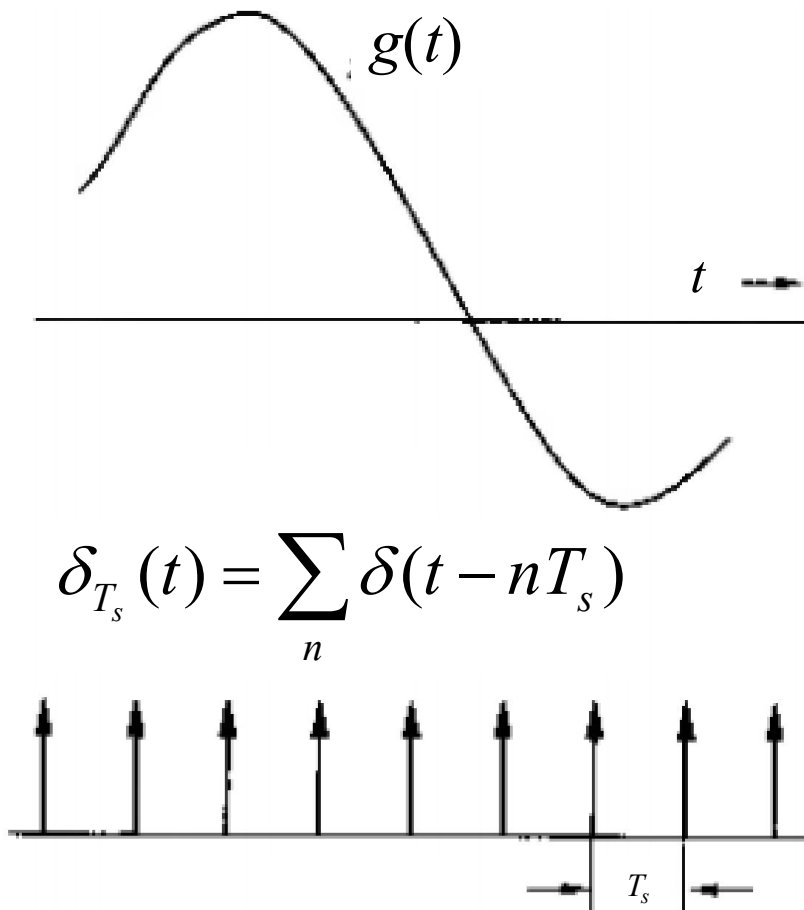
Multiply $g(t)$ by impulse train $\delta_{T_s}(t)$

where sampling period $T_s = 1/f_s$

The result is $\bar{g}(t)$

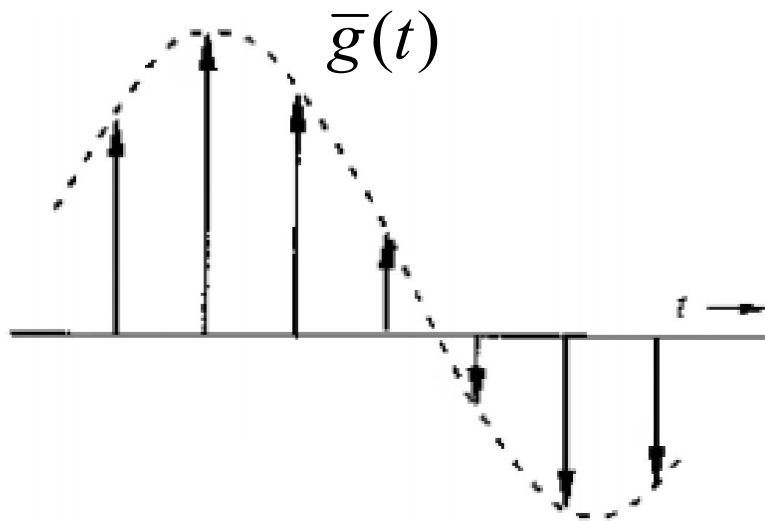
Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s



Sampling Theorem

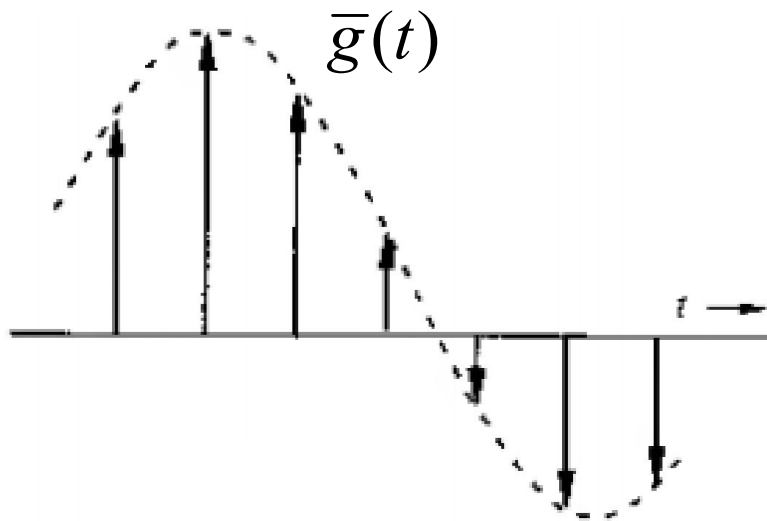
Uniform sampling $g(t)$ at a rate f_s



- $\bar{g}(t)$ consists of impulses uniformly spaced T_s seconds apart
- n th impulse is weighted by $g(nT_s)$, the value of $g(t)$ at $t = nT_s$

Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s

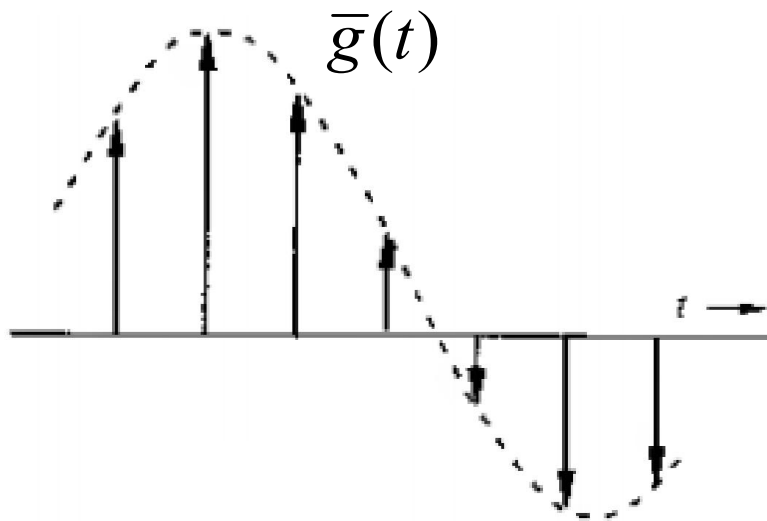


$\bar{g}(t)$ is multiplication of $g(t)$ and impulse train $\delta_{T_s}(t)$ and can be written as

$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$

Sampling Theorem

Uniform sampling $g(t)$ at a rate f_s



$$\phi(t)\delta(t) = \phi(0)\delta(t)$$

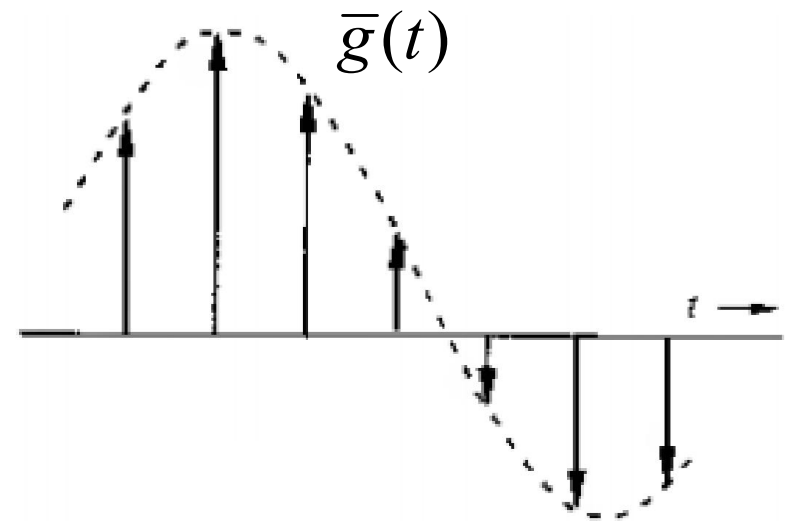
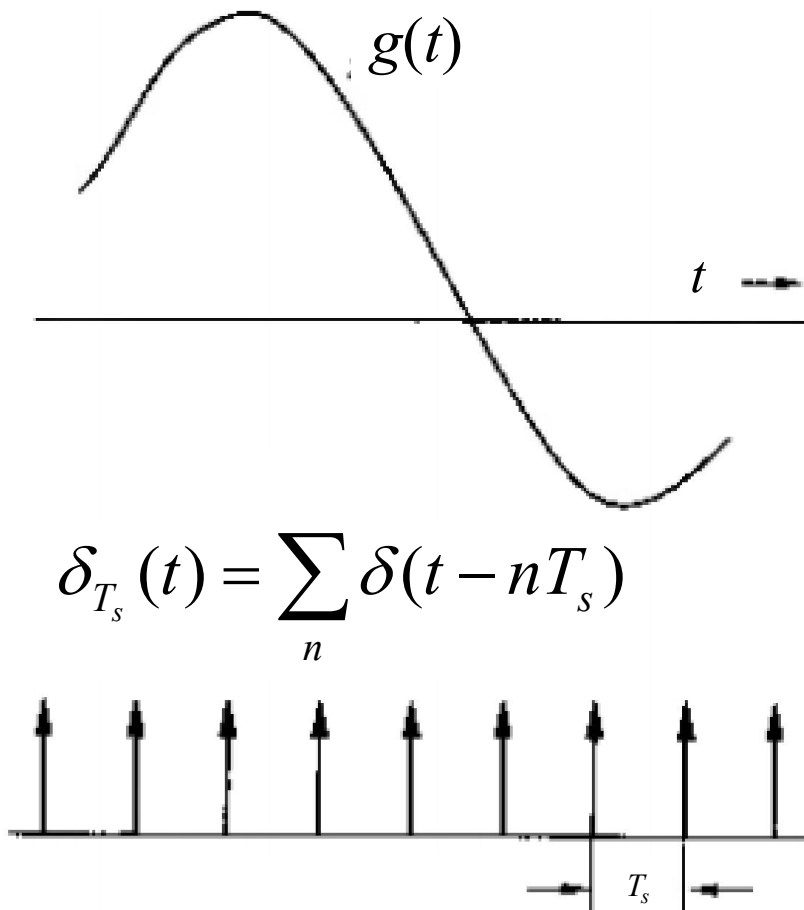
$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$

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Sampling Theorem

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$

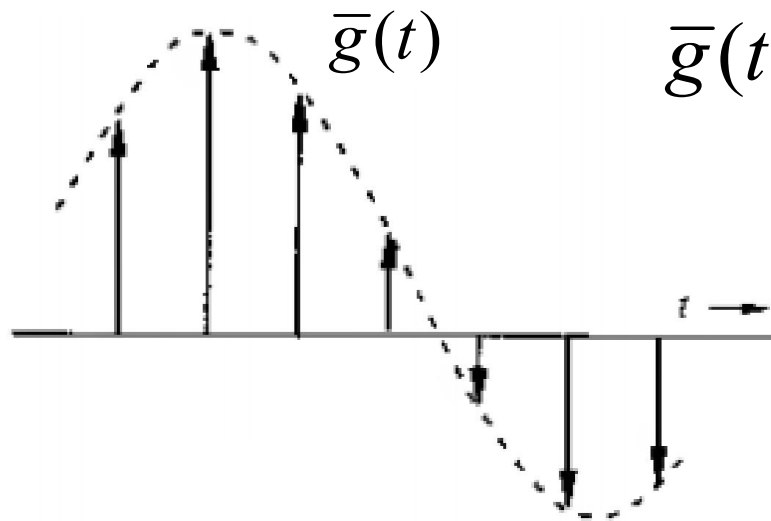


Alternate Representation

$$\delta_{T_s}(t) = \frac{1}{T_s} \sum_n e^{jn2\pi f_s t}$$

Sampling Theorem

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$



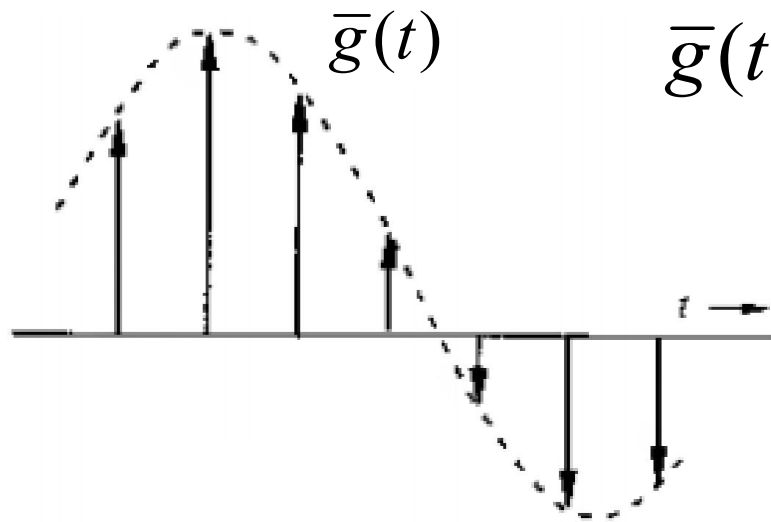
$$\begin{aligned}\bar{g}(t) &= g(t)\delta_{T_s}(t) = \frac{1}{T_s} \sum_n g(t)e^{jn2\pi f_s t} \\ &= \frac{1}{T_s} \sum_n g(t)e^{j2\pi n f_s t}\end{aligned}$$

$\bar{g}(t)$ is multiplication of $g(t)$ and impulse train $\delta_{T_s}(t)$ and can be written as

$$\bar{g}(t) = g(t)\delta_{T_s}(t) = \sum_n g(nT_s)\delta(t - nT_s)$$

Sampling Theorem

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$



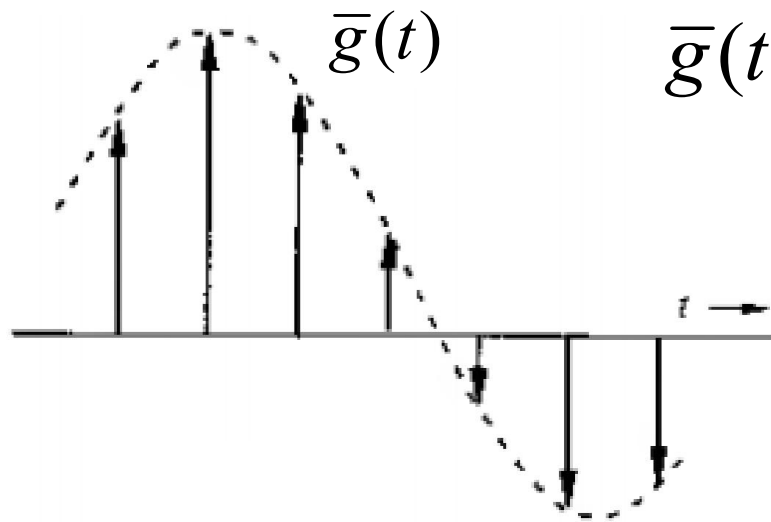
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FT of $\bar{g}(t)$ is defined by

$$\bar{G}(f) = \mathfrak{F}[\bar{g}(t)] = \frac{1}{T_s} \sum_n G(f - nf_s)$$

Sampling Theorem

Uniform sampling $g(t)$ at a rate $f_s = 1/T_s$



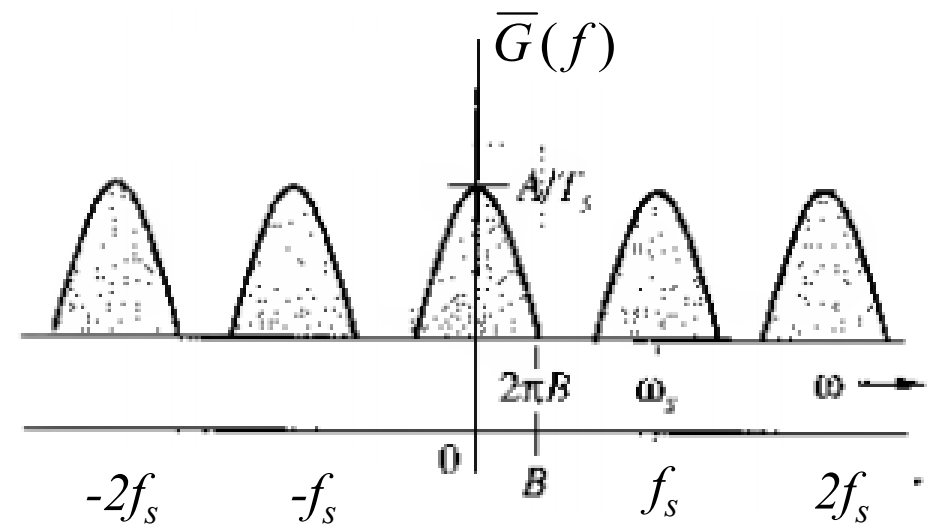
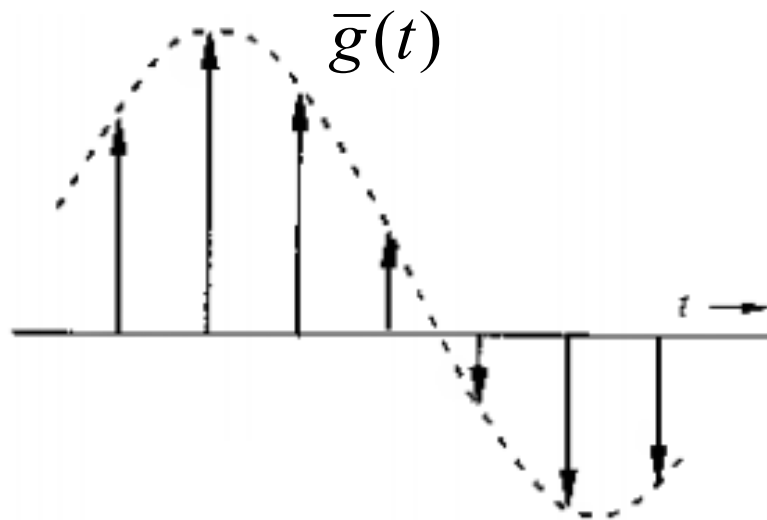
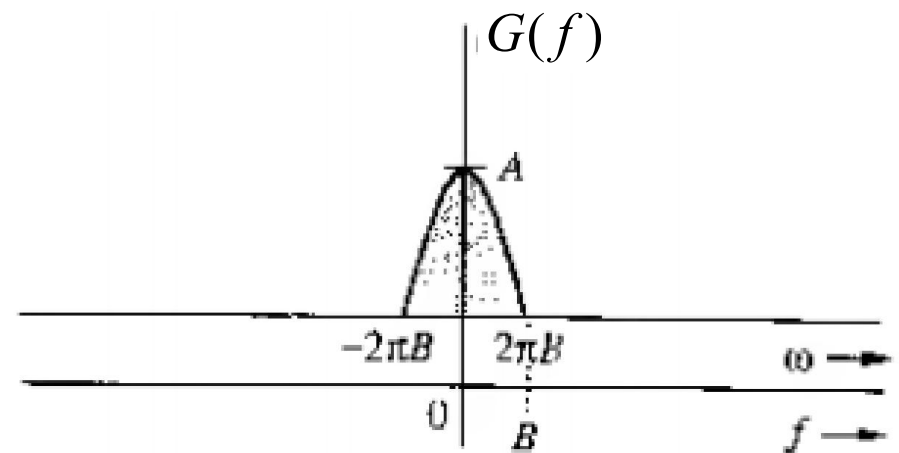
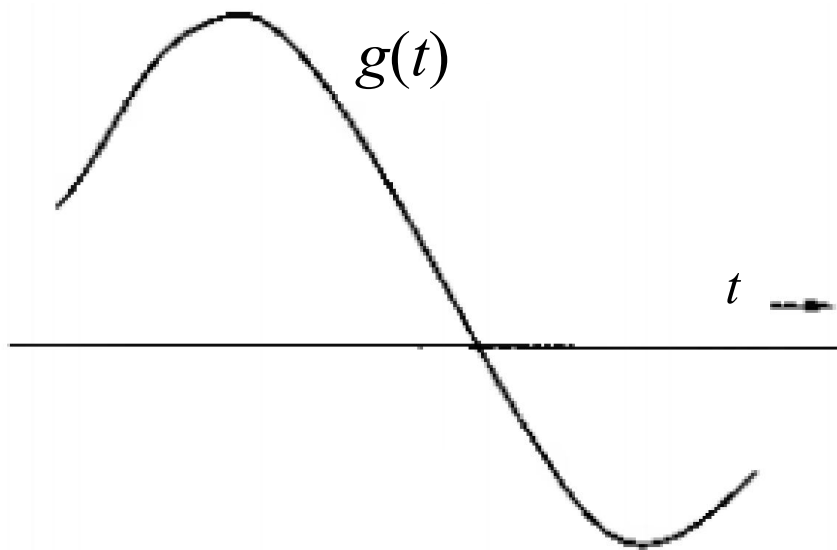
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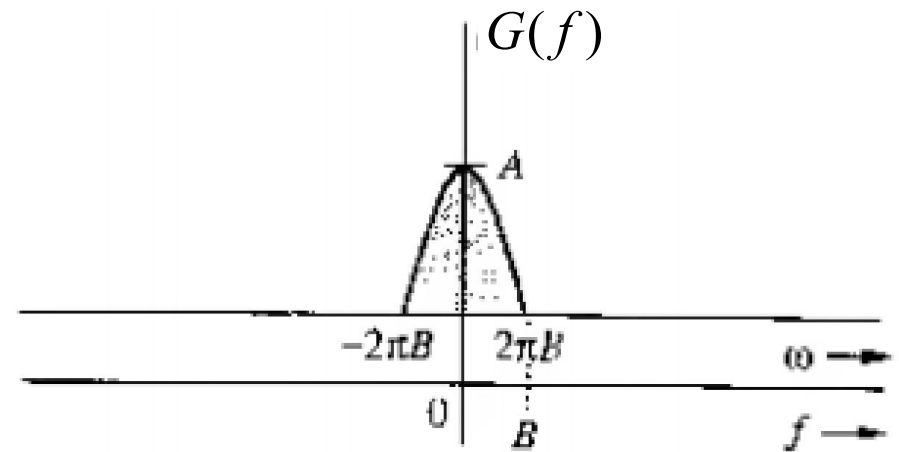
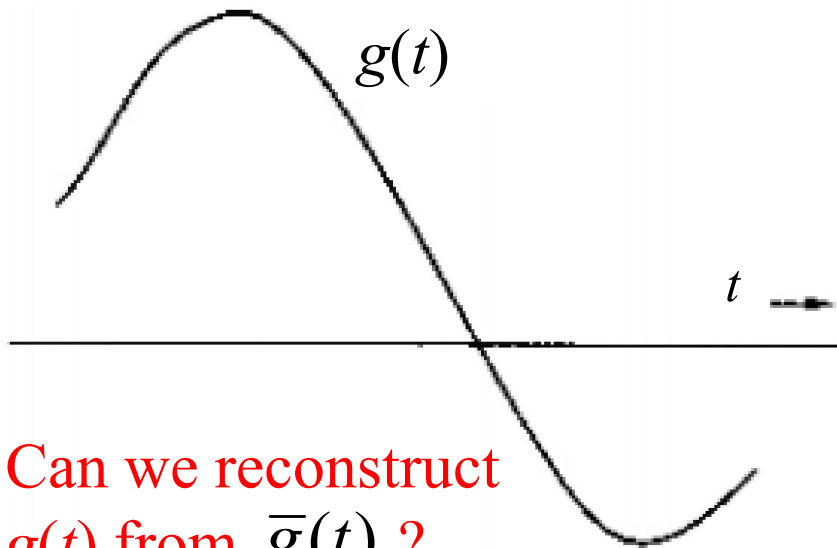
$$\bar{G}(f) = \mathfrak{F}[\bar{g}(t)] = \frac{1}{T_s} \sum_n G(f - nf_s)$$

$\bar{G}(f)$ consists of $G(f)$
scaled by f_s repeated
periodically with
period f_s

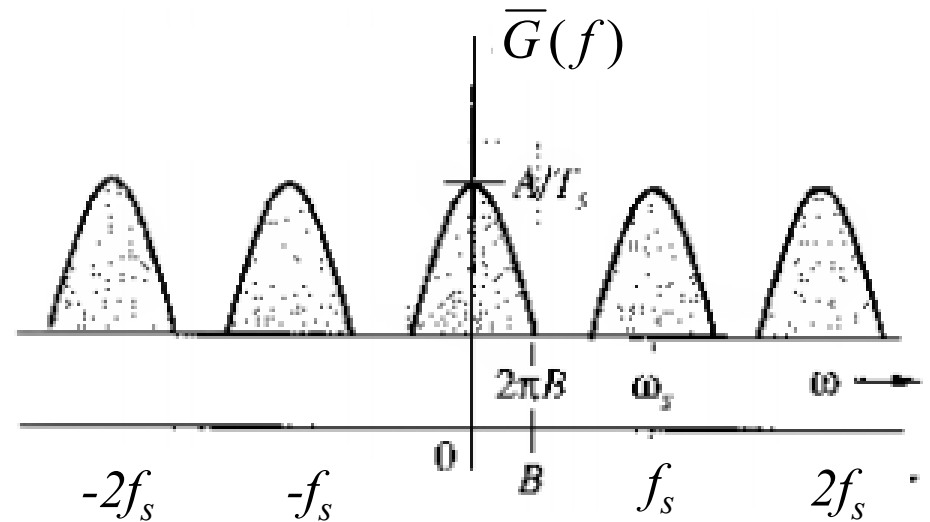
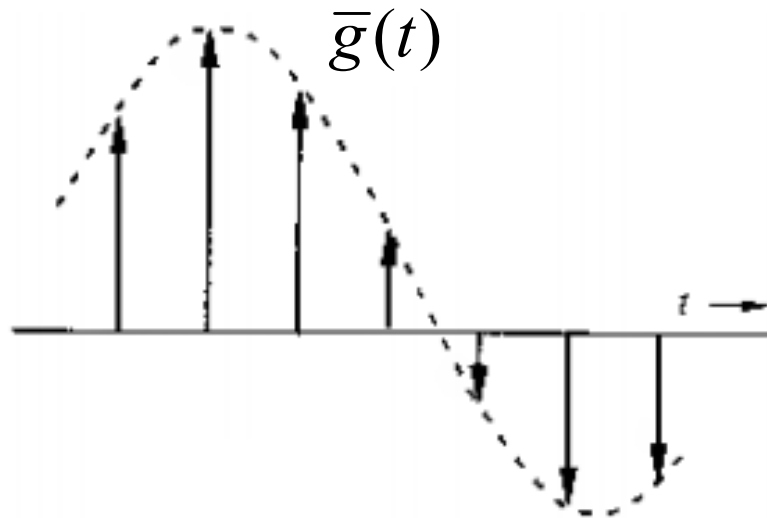
Sampling Theorem



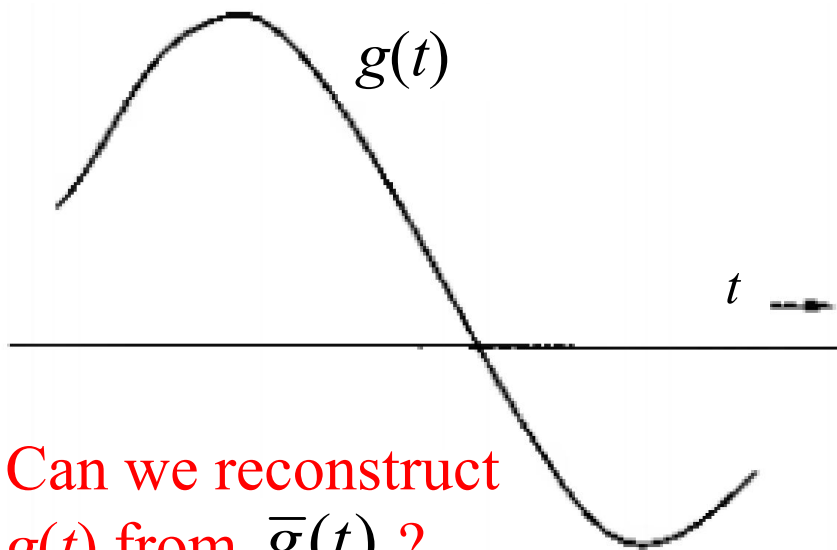
Sampling Theorem



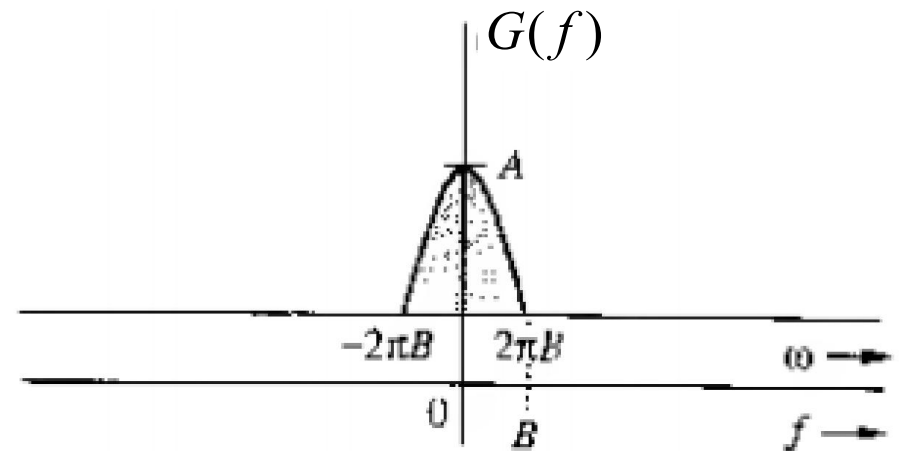
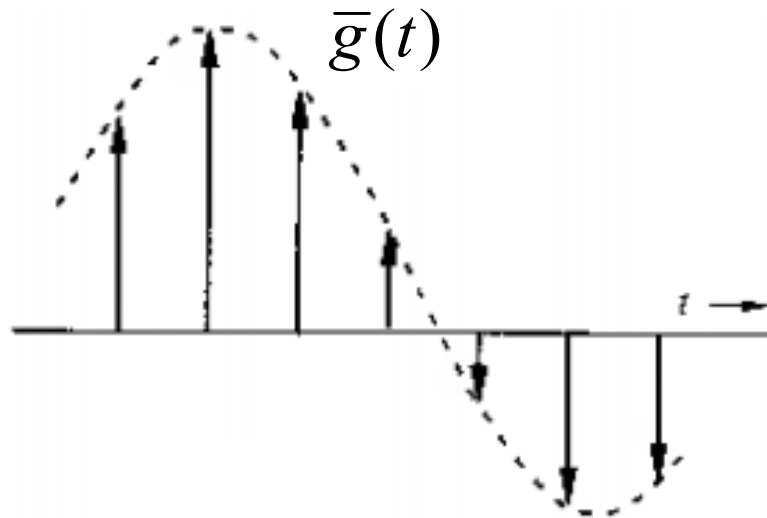
Can we reconstruct $g(t)$ from $\bar{g}(t)$?



Sampling Theorem



Can we reconstruct $g(t)$ from $\bar{g}(t)$?



Possible, if we can
reconstruct $G(f)$
from $\bar{G}(f)$

